

A Statistical Toolkit for Linguistic Typology: Comparing constraints

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Plan (tentative)

- Session 1: testing cross-linguistic generalizations
- Session 2: creating semantic maps
- **Session 3:**
 - comparing constraints
 - discussing your own ideas

Course materials

See the slides, data and code at
<https://github.com/levshina/Frejus2026>

(Session 3)

OR in the shared directory (see email)

Outline

- Theoretical background: Communicative efficiency
- Conditional inference trees and random forests
- Illustration: Causative constructions of the world
- Hands-on case study with R: Order of A and P in three languages
- Discussion of your own research ideas, Q&A

What is efficiency?

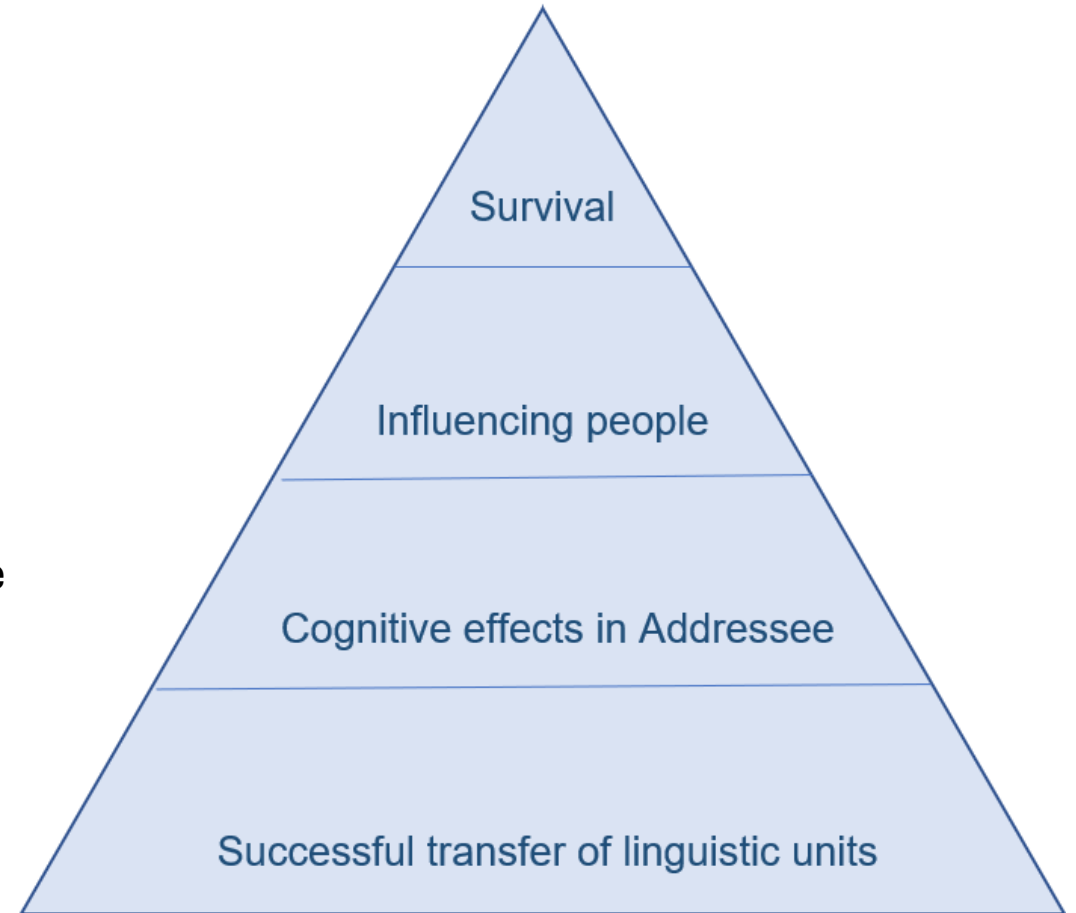
- Efficiency means minimization of a cost-to-benefit ratio. Being efficient means not spending more effort than necessary in order to achieve something.
- Living organisms try to save effort:
 - Penguins waddle because it conserves energy in comparison with walking.
 - Professional runners position their heels in such a way as to lower metabolic energy consumption.



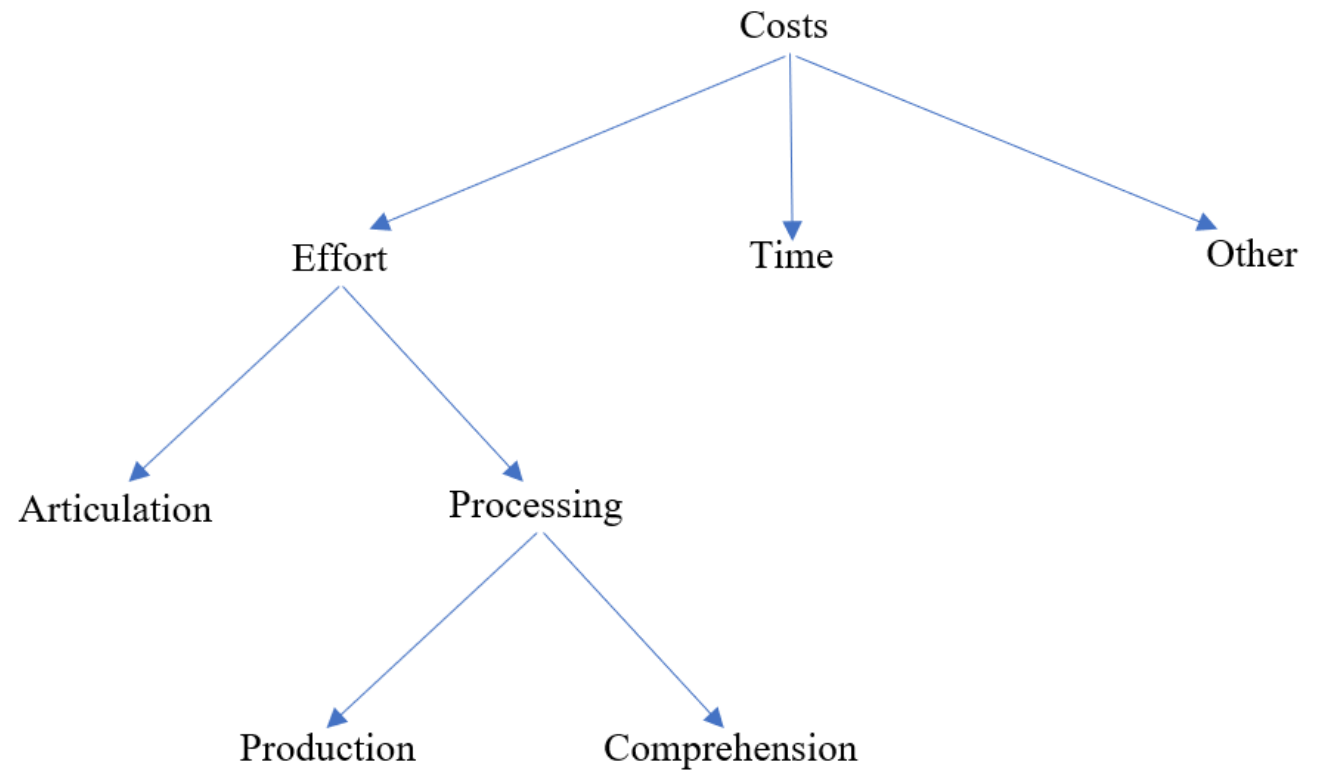
Benefits of linguistic communication

Jakobson's functions of language

- *referential*: describes a situation, object or mental state.
- *poetic*: "the message for its own sake"
- *emotive*: give information about the speaker's internal state
- *conative*: engages the Addressee directly
- *phatic*: language for the sake of interaction
- *metalinguistic*: the use of language



Costs of linguistic communication



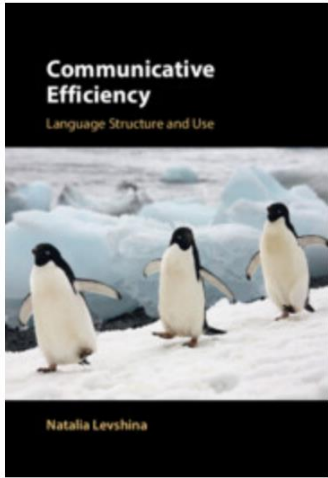
Communicative Efficiency

Language Structure and Use

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How to be communicatively
efficient?

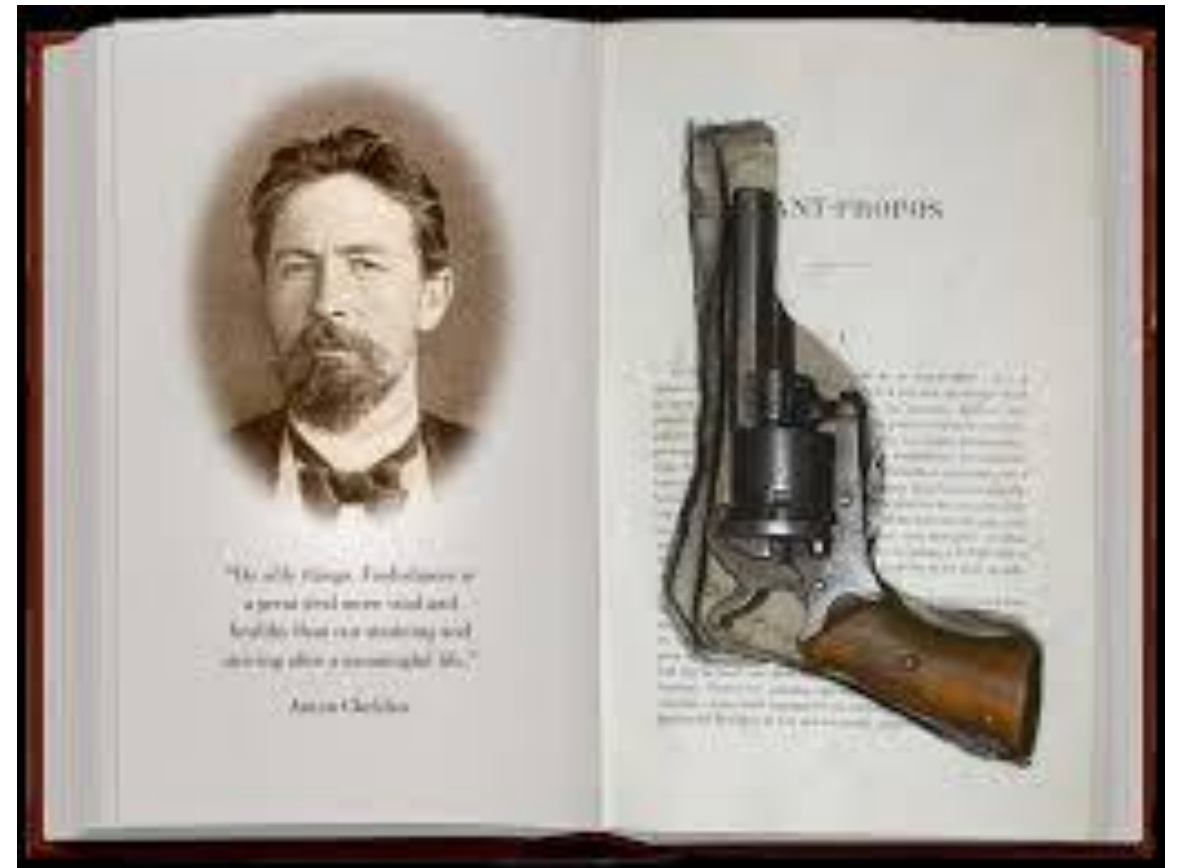
Principles of efficient communication

Principles of efficient communication

- **Positive correlation between benefits and costs**
 - Don't spend effort and time on useless information
 - Extra costs should be justified by extra benefits

*If there is a gun hanging on the wall in
the first act, it must fire in the last.*

Anton Chekhov

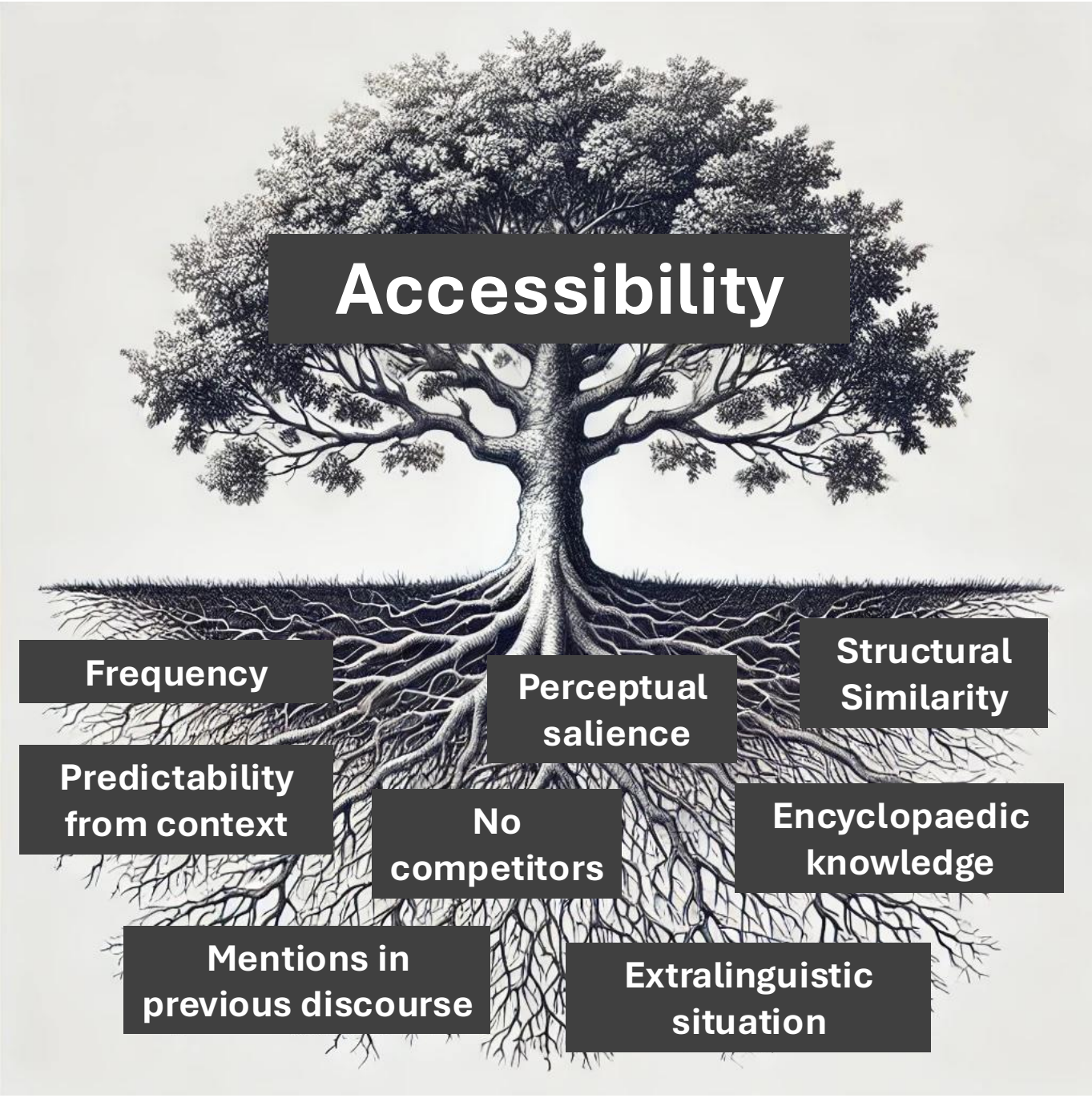


Positive correlation between benefits and costs

- Have you heard the news? ...
 - Bill was robbed yesterday. He's still under shock. Poor guy!
 - A tourist was robbed **of his Rolex watch**. He should have left it in a hotel safe!
 - A tourist was robbed **near the police station** yesterday. These criminals are becoming more and more daring!

Principles of efficient communication

- **Positive correlation between benefits and costs**
 - Don't spend effort and time on useless information
 - Extra costs should be justified by extra benefits
- **Negative correlation between accessibility and costs**
 - Spend less effort and time on more accessible information
 - Spend more effort and time on less accessible information



Accessibility

Frequency

**Perceptual
salience**

**Structural
Similarity**

**Predictability
from context**

**No
competitors**

**Encyclopaedic
knowledge**

**Mentions in
previous discourse**

**Extralinguistic
situation**

Principles of efficient communication

- **Positive correlation between benefits and costs**
 - Don't spend effort and time on useless information
 - Extra costs should be justified by extra benefits
- **Negative correlation between accessibility and costs**
 - Spend less effort and time on more accessible information
 - Spend more effort and time on less accessible information
- **Maximization of accessibility**
 - Maximize accessibility of information at every point in discourse

Maximization of accessibility

- Producing more accessible constituents earlier ("Easy first", Given-before-new, etc.)
- Placing (more) related meaningful elements closer to each other:
 - Minimization of dependency distances
 - Minimization of syntactic and semantic domains
 - Information locality principle
 - Cross-linguistic rules for the order of morphemes, etc.
 - Etc.

Cross-linguistic correlations between formal length and accessibility

- Although languages tend to observe this correlation, they vary in the role of individual factors and cut-off points.
 - DOM: animate/given/1st or 2nd person/pronominal objects are more likely to be formally marked than inanimate/new/3rd person/nominal objects (see Session 1)
 - Zero anaphora vs. explicit referents: Although it is agreed that zero anaphora is boosted by referent accessibility (Ariel 1990), variation is language-specific and depends on a multitude of factors that vary across languages: discourse continuity, antecedent distance, contrast/focus, inferability, grammatical role, verb indexing, word order...
- How to model this variability and find order in this chaos?

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Classification and Regression Trees (CART)

- Allow to model the relationships between a response variable and different predictors
 - E.g. Income depending on education, gender, work experience, etc.
 - Reaction times based on word length and frequency
 - The use of near-synonyms depending on contextual factors
- One can have categorical response (classification) or numeric (regression) and any kind of predictors (numeric and categorical).

Recursive binary partitioning

- Step 1. The algorithm starts with the entire data set and picks up the predictor the most strongly associated with the response.
- Step 2. The algorithm splits the data into two data sets (binary) according to the values of that predictor (partitioning).
- Steps 1 and 2 are repeated again and again until a certain criterion is met (recursive).

Conditional inference trees

- A specific algorithm for CART
- Works well with correlated predictors
- Can deal with missing values in predictors
- the splits are made in the predictor whose association/correlation with the response yields the smallest p-value
- stops when no more associations/correlations between the response and predictors smaller than the significance level (0.05) can be found. → No need to prune the tree (to avoid overfitting)

Random forests

- **Ensembles** of many decision trees
- Like ordinary Random forest models, they build many trees on resampled data.
- Predictions are obtained by aggregating trees (majority vote for classification, averaging for regression).
- They return **variable importance scores** that enable us to compare the ranking of different factors across languages.

More on trees and forests

Levshina, N. (2020). Conditional inference trees and random forests. In M. Paquot & S. Th. Gries (Eds.), *Practical Handbook of Corpus Linguistics* (pp. 611–643). Springer.

PDF: https://levshina.github.io/docs/Levshina2020_Trees&Forests.pdf

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A linguistic universal

“[T]he kind of formal distinction found across languages is identical: the continuum from analytic via morphological to lexical causative correlates with the continuum from less direct to more direct causation.”

(Comrie 1981: 165)

The causative continuum

Lexical <> Morphological <> Analytic (Periphrastic)

e.g. *kill, break*

e.g. Turkish *öldür*- “kill”
from *öl*- “die”

e.g. *cause* *X* to die,
make *X* disappear



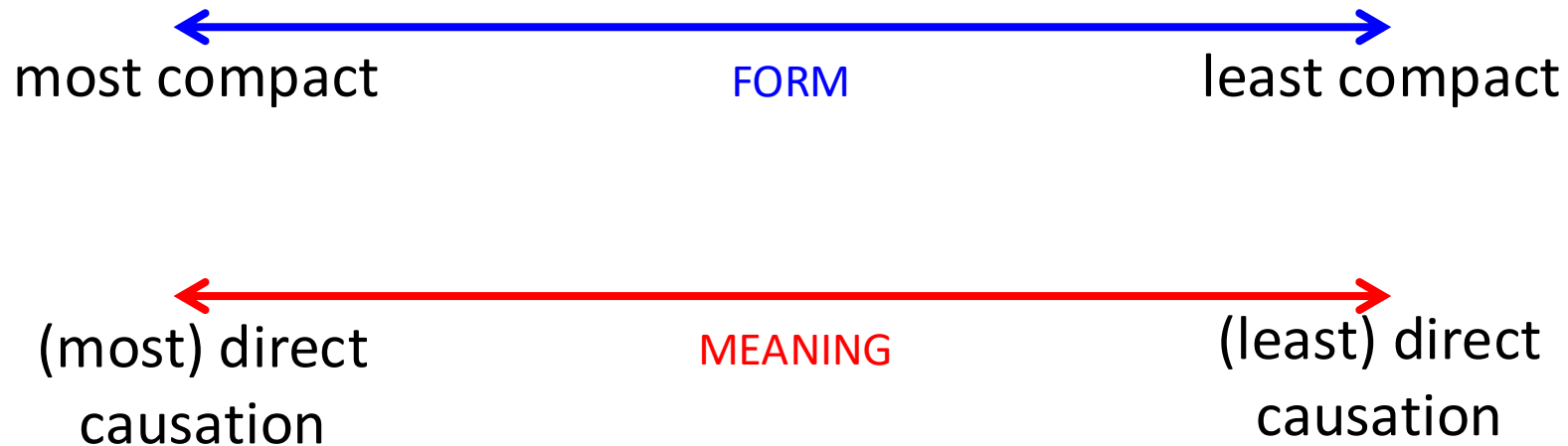
The causative continuum

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e.g. *cause* *X to die*,
make *X disappear*



Comrie 1981, Dixon 2000, Shibatani & Pardeshi 2002, etc.

Example

- Harry Potter **raised** the glass. [Causer acts directly]



- Harry Potter **caused** the glass **to rise**. [Causer acts indirectly]



Iconicity of cohesion

- “Meanings that belong together more closely semantically are expressed by more cohesive forms” (Haspelmath 2008: 2)
- If cause and effect are closely integrated semantically, the elements that express them will be formally integrated, too.

Dixon's (2000) multifactorial account

	More compact causative forms	Less compact causative forms
1.	non-causal verb describing a state	non-causal verb describing an action
2.	intransitive (or intransitive and simple transitive) non-causal verb	transitive (ditransitive) non-causal verb
3.	Causee lacking control	Causee having control
4.	Causee willing ('let')	Causee unwilling ('make')
5.	Causee partially affected	Causee fully affected
6.	Causer acts directly	Causer acts indirectly
7.	Causer achieves the result intentionally	Causer achieves the result accidentally
8.	causation occurring naturally	causation occurring with effort

Is Dixon's multivariate account supported by corpus data?

Languages

Language	Genus	Family
Chinese	Chinese	Sino-Tibetan
Finnish	Finnic	Uralic
French	Romance	Indo-European
Hebrew	Semitic	Afro-Asiatic
Indonesian	Malayo-Sumbawan	Austronesian
Japanese	Japanese	Japanese
Russian	Slavic	Indo-European
Thai	Kam-Tai	Tai-Kadai
Turkish	Turkic	Altaic
Vietnamese	Viet-Muong	Austro-Asiatic

Data set

- 344 causative situations found in the English segment of the ParTy corpus*
- Coded for various semantic and syntactic variables (based on Dixon's parameters): ONLY the English version!
- Translations in the 10 languages are found and coded into 3 types of constructions (Analytic, Morphological or Lexical)

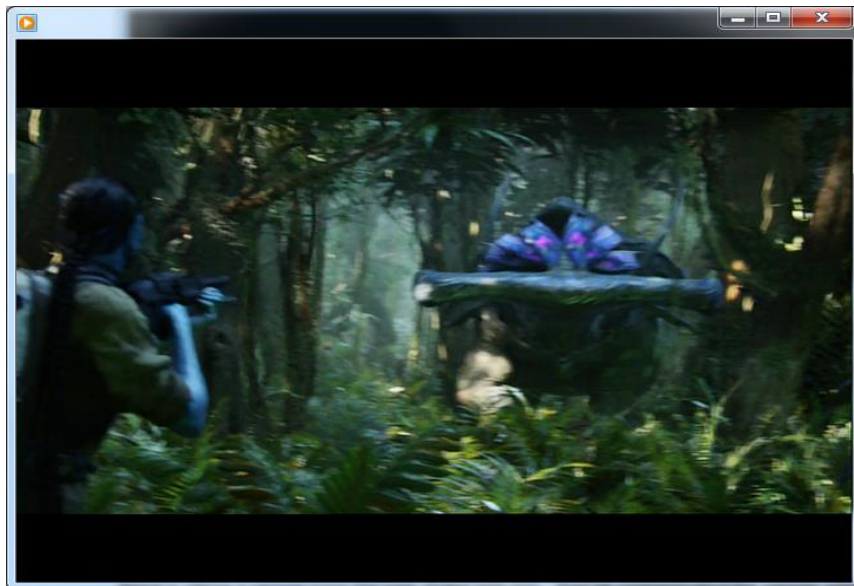
* <https://github.com/levshina/ParTy-1.0>

Variable	Values	Examples
Semantics of the caused event	‘State’ ‘Action’	John killed Harry. The teacher had the students ask questions
Number of main participants	‘2’ ‘3’	John killed Harry. John made Harry kill Mary.
Controlling Causee	‘Yes’ ‘No’	The teacher had the students ask questions. John killed Harry.
Causee acting willingly	‘Yes’ ‘No’ ‘Undef’	The teacher let the students leave earlier. The minister made the journalists wait for him.
Making or letting	‘Make’ ‘Let’	John killed Harry. The teacher let the students leave earlier.
Causar acting directly	‘Yes’ ‘No’	John broke Harry’s arm during the fight. The teacher had the students ask questions.
Causar acting intentionally	‘Yes’ ‘No’	The thieves broke the window to get in. Oops, I’ve broken your Ming vase.
Causar acting forcefully	‘Yes’ ‘No’	John forced Harry to kill Mary. The teacher had students ask questions.
Causar involved in caused event	‘Yes’ ‘No’	Bring your friends! John killed Harry.
Semantics of Causar	‘Anim’ ‘Inanim’	John killed Harry. Your haircut made me think of a bird’s nest.
Semantics of Causee	‘Anim’ ‘Inanim’	John killed Harry. I broke the vase.

Example from *Avatar*

Original

- ENG: *Don't shoot, you'll piss him off.*



Translations

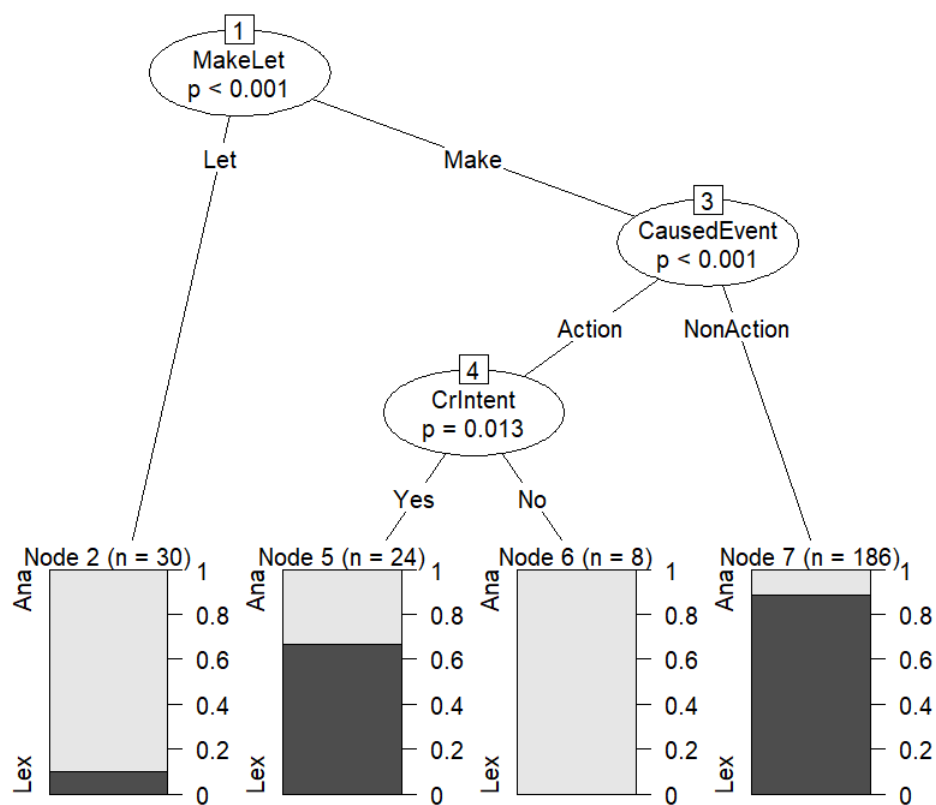
- FRA: *Ne tirez pas. Vous allez l'**énerv**er.* (Lexical)
- TUR: *Ateş etme. Ateş etme. Onu kız**dır**acaksın.* (Morphological, from *kızmek* 'become angry').
- VIE: *Đừng bắn. Cậu sẽ **làm** nó **nổi** **điên** đó.* (Analytic)

Examples of constructions

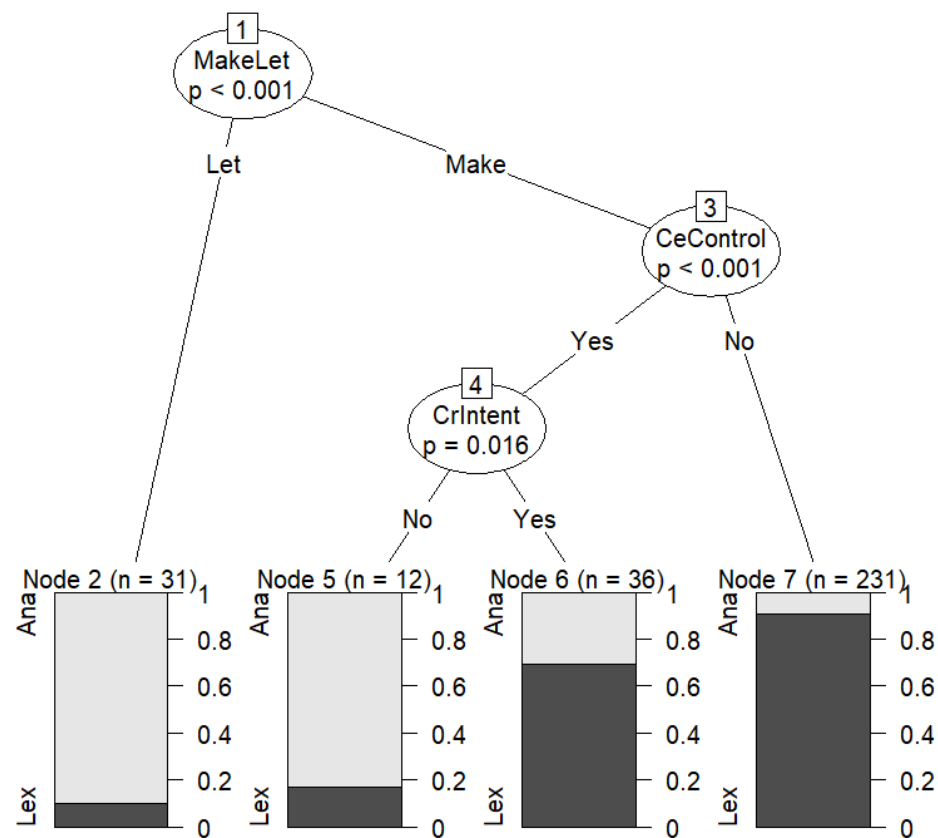
Language	Lexical	Morphological	Analytic
Chinese	sha “kill”	-	ràng “let, make” + Pred
Finnish	tappaa “kill”	odotu-tt-aa “make wait”	antaa “give” + V1
French	tuer “kill”	-	faire + Vinf
Hebrew	harag “kill” pa'al	hotsi “take out” hip'h'il	natan “give” + le-Vinf
Indonesian	mem-bunuh “kill”	meng-ingat-kan “remind”	membuat “make” + Pred
Japanese	korosu “kill”	ikar-ase-ru “make angry”	V_te + morau “get”
Russian	ubit' “kill”	-	zastavit' + Vinf
Thai	kaa “kill	-	tham hai “do give” + Pred
Turkish	açmak “open”	öl-dür- “kill”	V_mA_DAT + izin ver- “allow”
Vietnamese	giết hại “kill”	-	làm “do” + Pred

Conditional inference trees: Analytic vs. Lexical

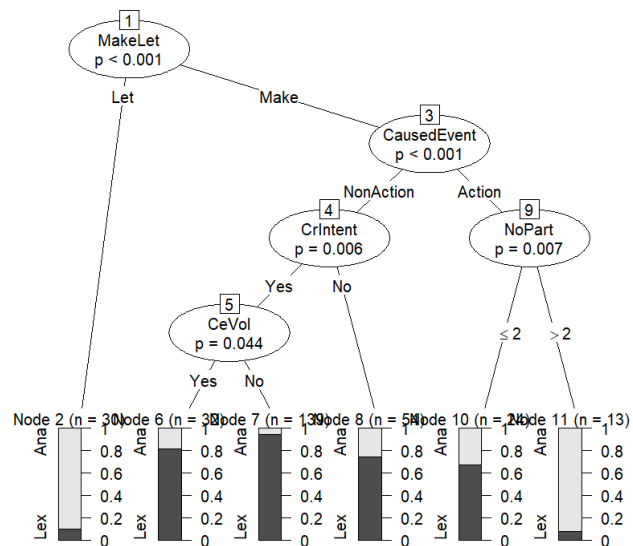
Finnish



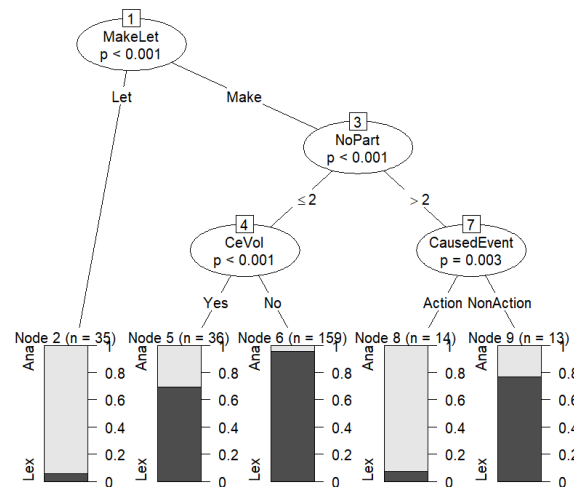
French



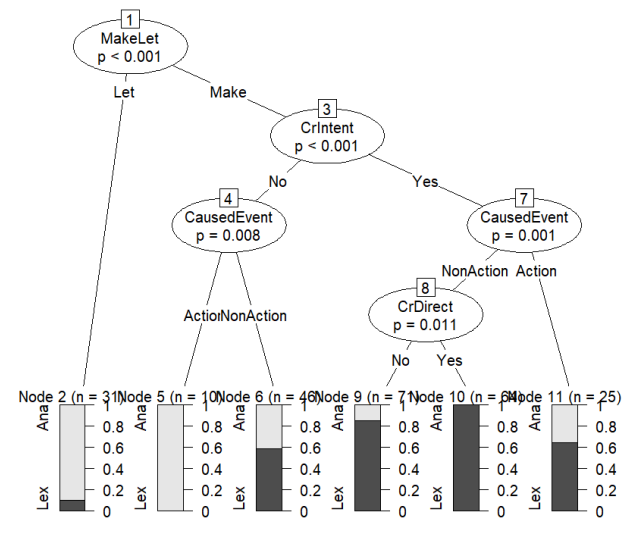
Chinese



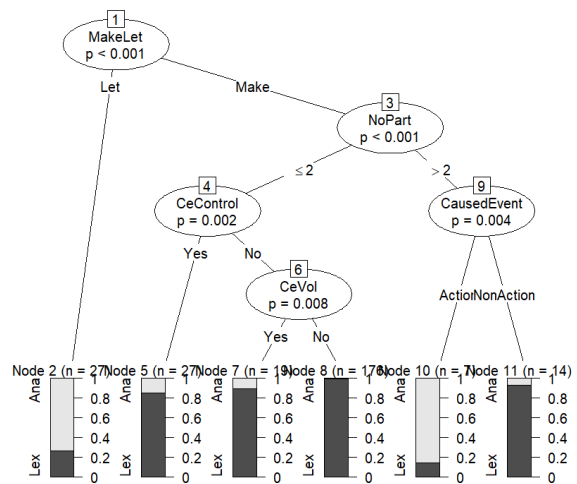
Hebrew



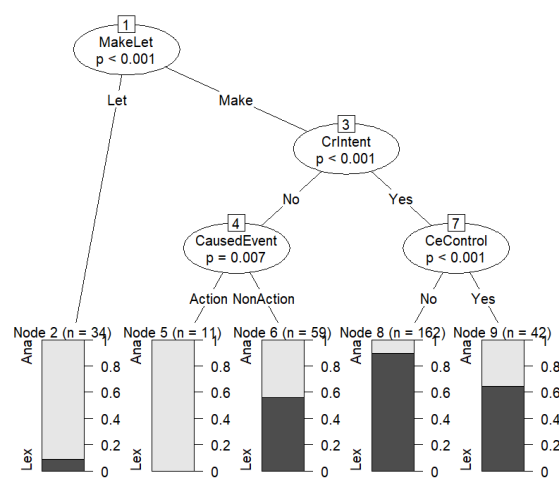
Indonesian



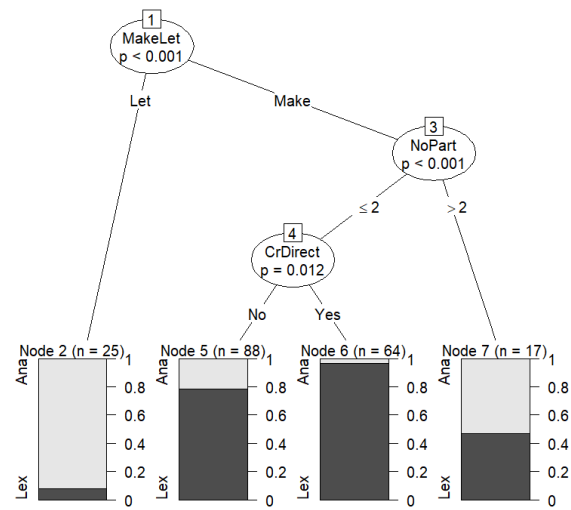
Russian



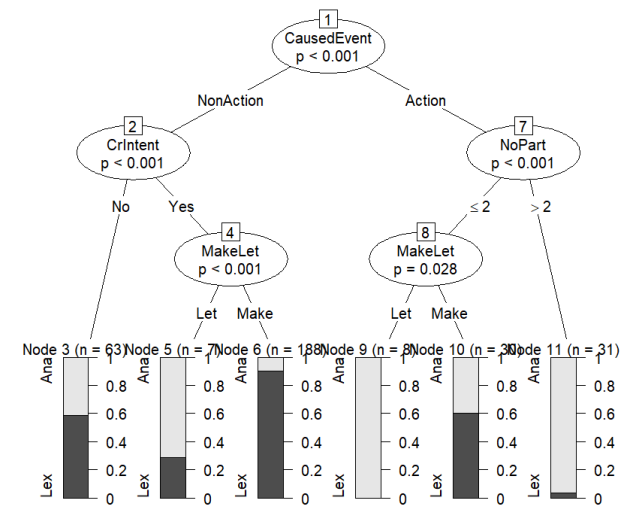
Thai



Turkish



Vietnamese

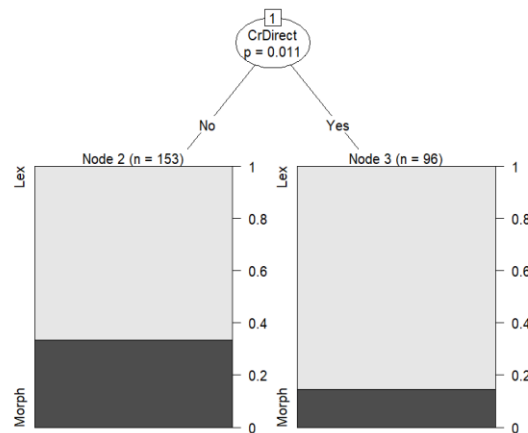


Features that distinguish Analytic from Lexical Causatives

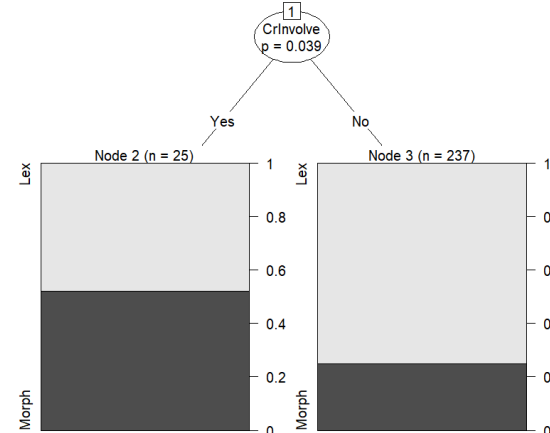
Language	CausedEvent	NoPart	CeControl	CeVol	MakeLet	CrDirect	CrIntent	CrForce	CrInvolved
Chinese	+	+		+	+		+		
Finnish	+				+		+		
French			+		+		+		
Hebrew	+	+		+	+				
Indonesian	+				+	+	+		
Russian	+	+	+	+	+				
Thai	+		+		+		+		
Turkish		+			+	+			
Vietnamese	+	+			+		+		

Conditional inference trees: Morphological vs. Lexical

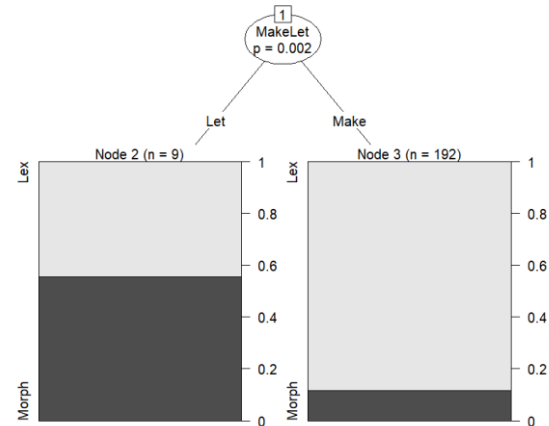
Finnish



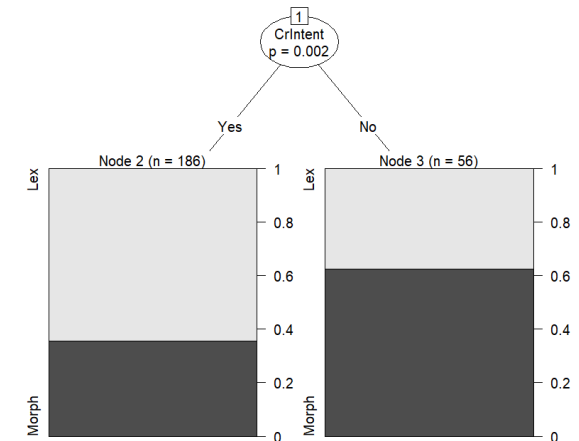
Hebrew



Japanese



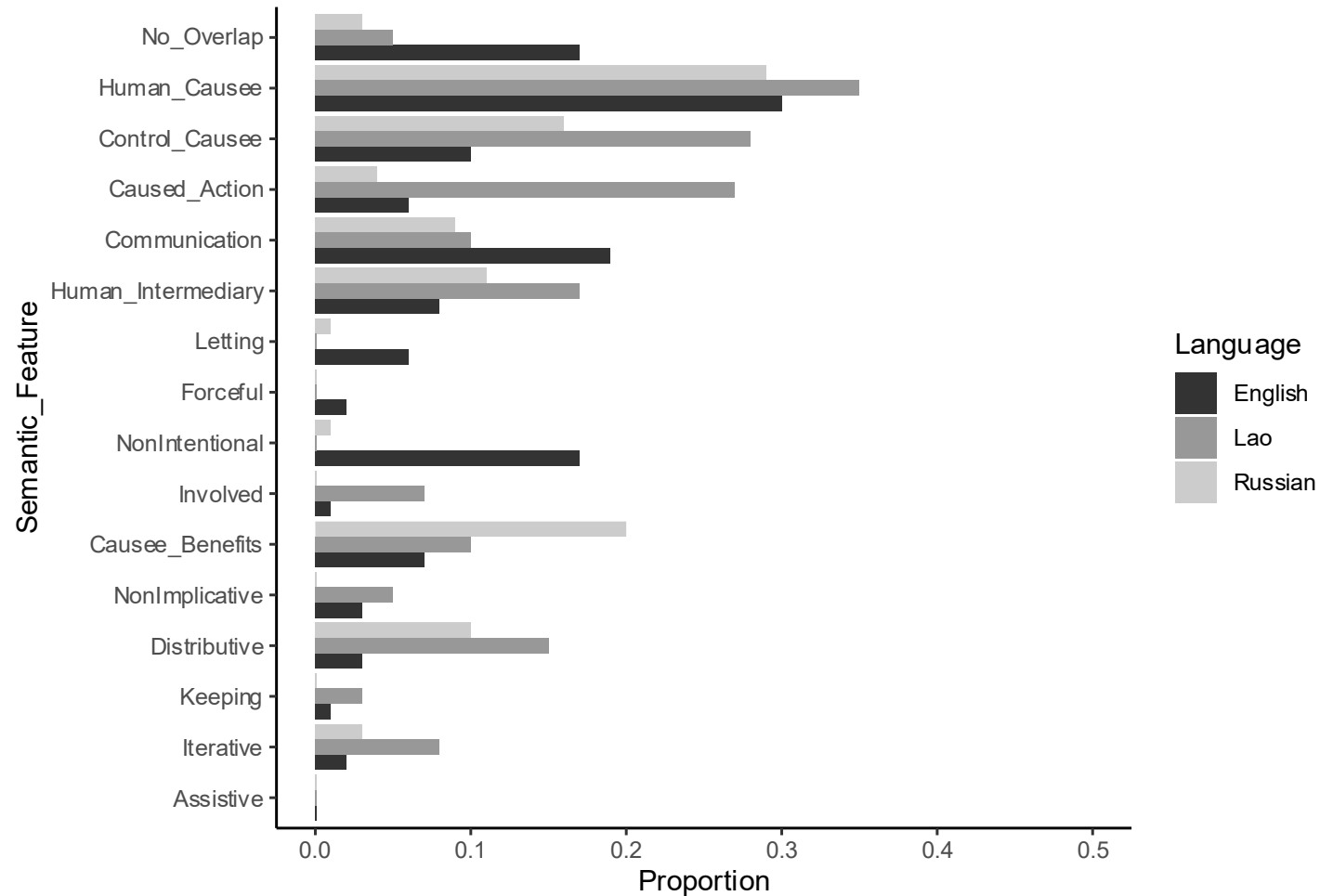
Turkish



Efficiency vs. iconicity

- The iconicity-based account (directness/indirectness) doesn't explain the multidimensional character of variation.
- A better explanation may be the principle of negative correlation between accessibility and costs:
 - Shorter constructions tend to refer to more frequent causative situations, and longer ones to less frequent causative situations (Haspelmath 2008).

Frequencies from three corpora



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A joint project

Rowena Garcia



Yevheniy Skyra



Anya Petukhova



Levelt Innovation Award

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Given vs. new information: competing motivations

- GIVEN before NEW as an "unmarked" order (Halliday 1967), also Behaghel (1932), Siewierska (1988), Gundel (1988), etc.
- MORE NEWSWORTHY before LESS NEWSWORTHY or TOPICAL
 - speakers of Cayuga (Iroquoian, the USA) and Ngandi (Gunwinyguan, Australia) prefer mentioning new or newsworthy constituents first (Chafe 1994; Mithun 1992)
 - Ute (Givón 1983), colloquial Mexican and Chilean Spanish (Bentivoglio 1983), Early Biblical Hebrew (Fox 1983), Nandi (Creider & Creider 1983): more easily identifiable topics are placed further to the right.
 - “Urgent information first”, “first things first” bias in spontaneous interaction (Givón 1983, Gundel 1988).

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A psycholinguistic perspective

- 'Easy First' as a source of word order flexibility
 - More frequent, shorter, less syntactically complex, given, more salient etc. words and phrases are more easily retrieved from long-term memory. When they come first, this maximizes fluency and allows more time to plan the more demanding elements.
- 'Plan Reuse' as a source of word order rigidity and priming effects:
 - Abstract sentence plans that have been recently used or that are frequent in general are easier to retrieve from long-term memory.
 - A source of overgeneralization of regular morphological patterns (e.g., holded vs. held)

Both make production
more efficient!

Research questions and expectations

- To what extent do speakers of German, Russian and Tagalog prefer the GIVEN-NEW order of A and P?
 - Expectation: very strongly
- What is the role of GIVEN-NEW in determining the order in comparison with other cues?
 - Expectation: more important than the role (A or P)
- Also, it has been claimed that language users use the canonical order (so-called “freezing”) in the absence of other cues to grammatical roles (Bouma 2011; Lee 2001). Is that true?
 - Expectation: systematically

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A and P in German

- Case marking (Nominative vs. Accusative) on masculine nouns
- Verb agreement with Subject sometimes helps to disambiguate
- Most popular orders: A before P, Verb-final or medial
- Example of variation:

Der
ART.M.**NOM**.SG

Hund
dog

verfolg-t
chase-PRS.3SG

den
ART.M.**ACC**.SG

Kater.
cat

Den
ART.M.**ACC**.SG

Kater
cat

verfolg-t
chase-PRS.3SG

der
ART.M.**NOM**.SG

Hund.
dog

'The dog chases the cat.'

A and P in Russian

- Case marking: Nominative vs. Accusative are distinguishable in many declension classes
- Verb agreement with Subject
- Most popular orders: A before P, Verb-middle or final
- Example of variation:

Lenin
Lenin.**NOM**.SG

potselova-l
kiss.PFV-PST.M.SG

Krupsk-uju.
Krupskaja-**ACC**.SG

Krupsk-uju
Krupskaja-**ACC**.SG

potselova-l-Ø
kiss.PFV-PST.M.SG

Lenin.
Lenin.**NOM**.SG

“Lenin kissed Krupskaja (his wife).”

A and P in Tagalog

- Case marking: *ang*= the "subject" and topic
- Voice marking on the verb (Actor vs. Patient voice) assigns the thematic role (A or P) of *ang*
- Most popular orders: verb-initial, A before P (in the Patient voice)
- Example of variation for "The frog tickled the bear":

S<um>i~sipa	<i>ang</i>	palaka	<i>ng</i>	oso.
<AV>IPFV~tickle	SBJ	frog	NSBJ	bear

S<um>i~sipa	<i>ng</i>	oso	<i>ang</i>	palaka.
<AV>IPFV~tickle	NSBJ	bear	SBJ	frog

S<in>i~sipa	<i>ng</i>	palaka	<i>ang</i>	oso.
<PV>IPFV~tickle	NSBJ	frog	SBJ	bear

S<in>i~sipa	<i>ang</i>	oso	<i>ng</i>	palaka.
<PV>IPFV~tickle	SBJ	bear	NSBJ	frog

Corpus data

Type of data	Language	Source	Number of transitive clauses
Adult-directed	German	Corpus of Spoken German IDS	308
	Russian	RNC, Zemskaia & Kapanadze 1978	300
	Tagalog	Garcia 2021	411
Child-directed	German	CHILDES (Leo)	309
	Russian	CHILDES (Protassova)	330
	Tagalog	Own corpus	399

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35 contextual variables

- Metainformation (speaker's ID, conversation ID, etc.)
- Properties of A and P
 - Semantics
 - Structure
 - Person
 - Length in syllables and in words
 - Discourse and situation status (see below)
- Properties of the verb, clause and utterance
 - Word order
 - Verb tense and aspect
 - Verb semantics
 - Utterance type
 - Mono- or ditransitive
 - Idiomaticity
- Translations and glosses

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Discourse status

Label	Meaning
Non-referential	The expression is not referential.
Mentioned_close	The referent is mentioned in the current or previous intonation unit.
Mentioned_far	The referent is mentioned further above than the previous intonation unit.
Mentioned	The referent is mentioned, but you are unsure how close.
Inferable_close	The referent is inferable from a referent mentioned in the current or previous intonation unit.
Inferable_far	The referent is inferable from a referent mentioned further above than the previous intonation unit.
Inferable	The referent is inferable, but you are not sure how close the second referent is.
New_familiar	The referent is neither mentioned, nor inferable, but the speaker assumes that it is in shared knowledge.
New_unfamiliar	The referent is neither mentioned, nor inferable, and not in shared knowledge.
New	The referent is neither mentioned nor inferable, but you are unsure if it is part of shared knowledge.
Unclear	When you really cannot decide

Situation status

Label	Meaning
Nonreferential	The expression is not referential.
Present_salient	The referent is present in the assumed extralinguistic context and salient.
Present_nonsalient	The referent is present in the assumed extralinguistic context and not salient.
Present	The referent is present in the assumed extralinguistic context, but you are not sure if it is salient or not.
Inferable_salient	The referent is inferable from a salient referent present in the assumed extralinguistic context.
Inferable_nonsalient	The referent is inferable from a non-salient referent present in the assumed extralinguistic context.
Inferable	The referent is inferable from a referent present in the assumed extralinguistic context, but you are not sure if the latter is inferable or not.
Absent_familiar	The referent is neither present, nor inferable, but the speaker assumes that it is part of shared knowledge.
Absent_unfamiliar	The referent is neither present, nor inferable, and not part of shared knowledge.
Absent	The referent is neither present, nor inferable, but you are not sure if it is part of shared knowledge.
Unclear	When you really cannot decide

Example of coding: German

<i>und</i>	<i>jetzt</i>	<i>darfst</i>	<i>du</i>	<i>wieder</i>	<i>eine</i>	<i>Karte</i>	<i>ziehen</i>
and	now	may-2SG	you.NOM.SG	again	a.ACC	card.ACC.SG	pull.INF

“and now you can take a card again”

- **A ("you")**: A_only caseform, 2nd Person, human, pronoun, 1 word, 1 syllable, Mentioned Far, Present Salient
- **P ("a card")**: ambiguous caseform, 3rd person, inanimate, NP, 2 words, 4 syllables, Mentioned Far, Present Salient
- General context: word order V1APV2, Causal verb (V2), Present tense, Declarative Sentence, No formal ambiguity, No idiomaticity, Not ditransitive

Example of coding: Russian

Да/ но куда родителей я дену? (RNC, Going Through Photos)

Da	no	kuda	roditelej	ja	denu
yes	but	where	parents.ACC/GEN.PL	1SG.NOM	put.IPFV.FUT.1.SG

“Yes, but where will I put my parents?”

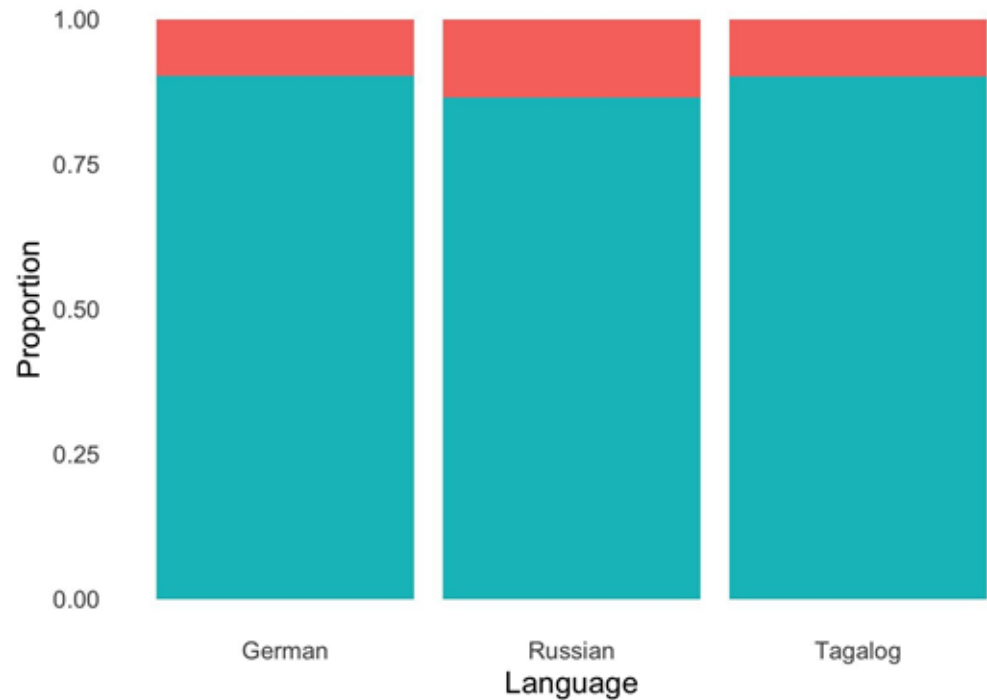
- **A ("I")**: A_only caseform, 1st person, human, pronoun, 1 word, 1 syllable, Mentioned far, Present Salient
- **P ("parents")**: P_only caseform, 3rd person, bare noun, 1 word, 4 syllables, Discourse-new, Absent Familiar
- General context: word order PAV, Causal verb, Future tense, Interrogative Sentence, No formal ambiguity, No idiomaticity, Not ditransitive

Examples of coding: Tagalog

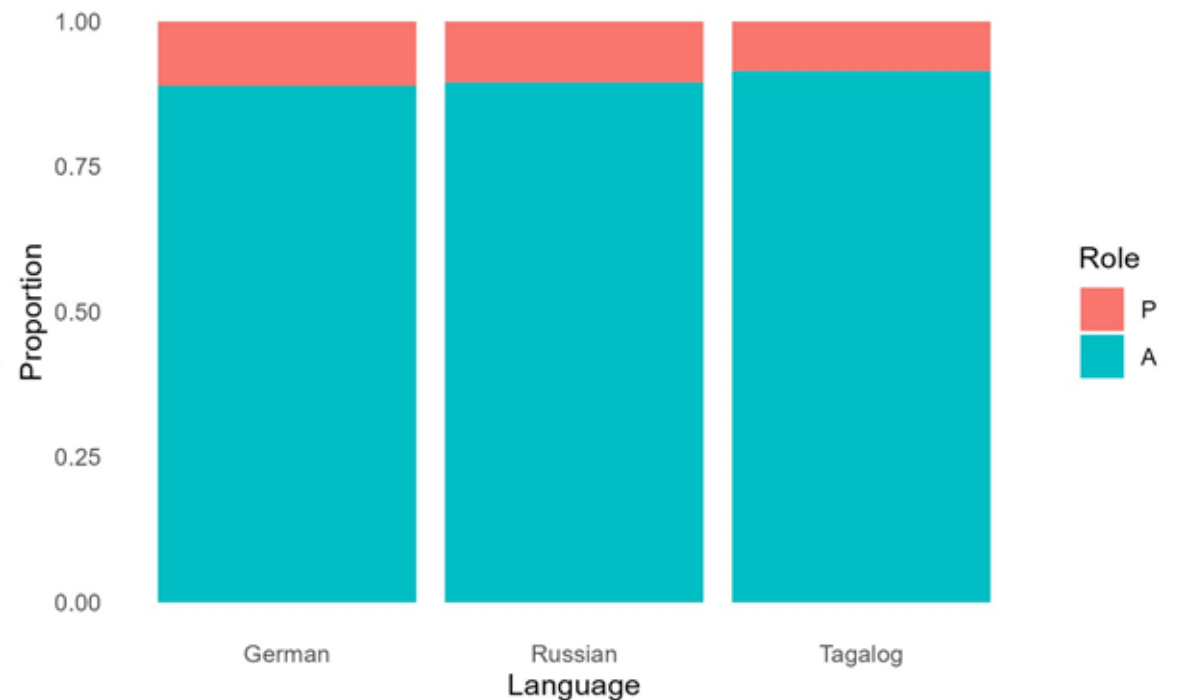
Nag-uwi *siya* *ng* *panga*
<AV>PFV.bring home 3SG.SBJ NSBJ jaw/collar (of fish)
“He/She brought home tuna collars”

- **A ("he/she")**: Ambiguous caseform, 3rd person, human, pronoun, 1 word, 1 syllable, Mentioned far, Present Salient
- **P ("tuna collars")**: Ambiguous caseform, 3rd person, inanimate, noun phrase, 2 words, 3 syllables, Mentioned close, Absent Familiar
- General context: word order VAP, Causal verb, Agent voice, Perfective aspect, Declarative Sentence, No formal ambiguity, No idiomaticity, Not ditransitive

First vs. Second Argument: Roles



Adult-directed speech



Child-directed speech

Levshina et al. Forthcoming

Create conditional inference trees for each language

- See the script in the Rmd/html file.

Conditional Random Forests

- Aggregation of 1000 trees
- Random sampling of predictors (5, 10 or 15) via **mtry**.
- Formula:

First/Second ~ Discourse_Status + Situation_Status + Length_Words +
Length_Syl + Semantics + Person + Structure + ...

Performance for adult-directed speech

- Accuracy (baseline = 0.5)

Language	Mtry = 5	Mtry = 10	Mtry = 15
German	0.901	0.901	0.901
Russian	0.873	0.877	0.879
Tagalog	0.890	0.898	0.897

- C-score, or Area under the ROC curve (baseline = 0.5)

Language	Mtry = 5	Mtry = 10	Mtry = 15
German	0.921	0.922	0.925
Russian	0.941	0.94	0.939
Tagalog	0.956	0.957	0.956

Performance for child-directed speech

- Accuracy (baseline = 0.5)

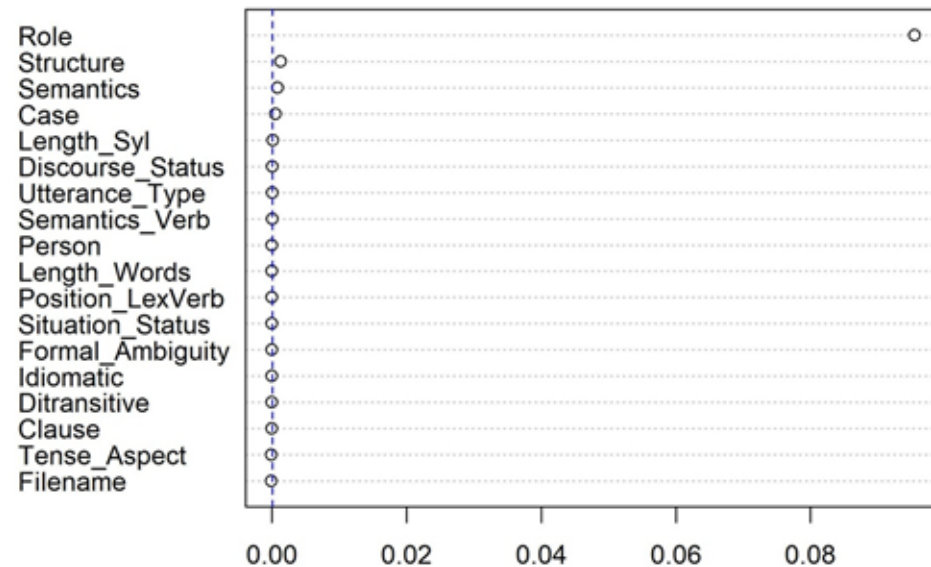
Language	Mtry = 5	Mtry = 10	Mtry = 15
German	0.889	0.889	0.889
Russian	0.895	0.895	0.895
Tagalog	0.94	0.946	0.946

- C-score, or Area under the ROC curve (baseline = 0.5)

Language	Mtry = 5	Mtry = 10	Mtry = 15
German	0.911	0.913	0.915
Russian	0.942	0.947	0.945
Tagalog	0.986	0.985	0.984

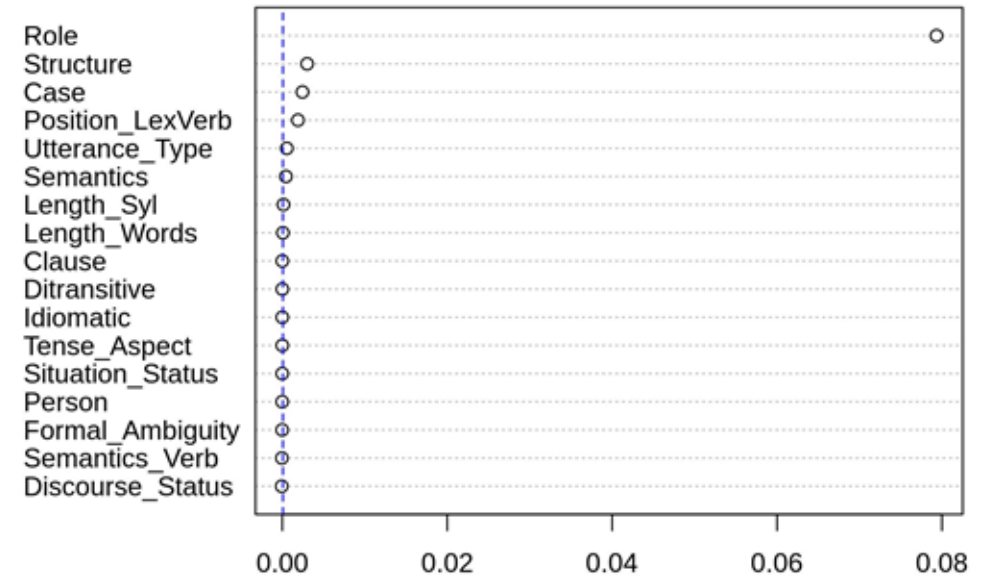
Conditional variable importance: German

German: cforest, Variable Importance, mtry = 10



Adult-directed speech

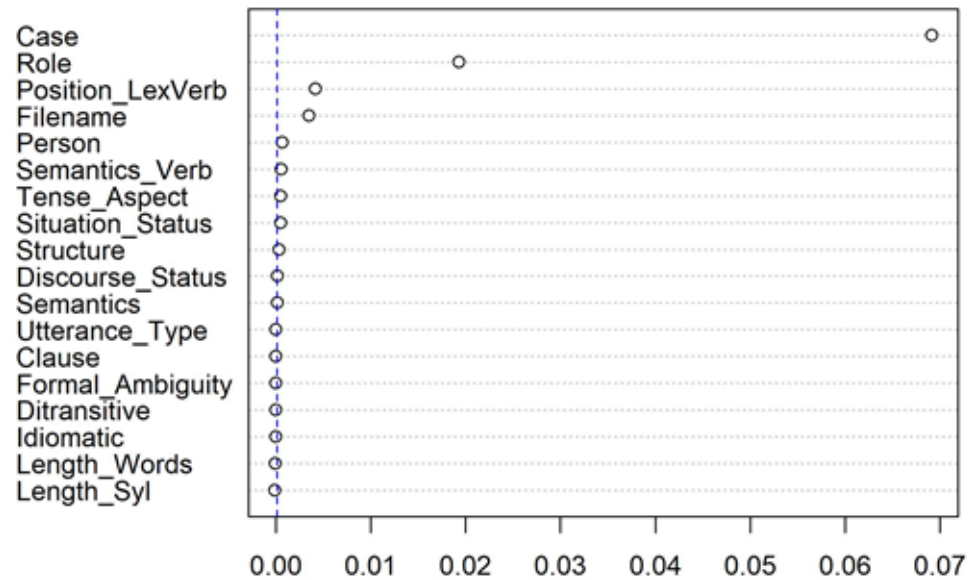
German CDS: cforest, Variable Importance, mtry = 10



Child-directed speech

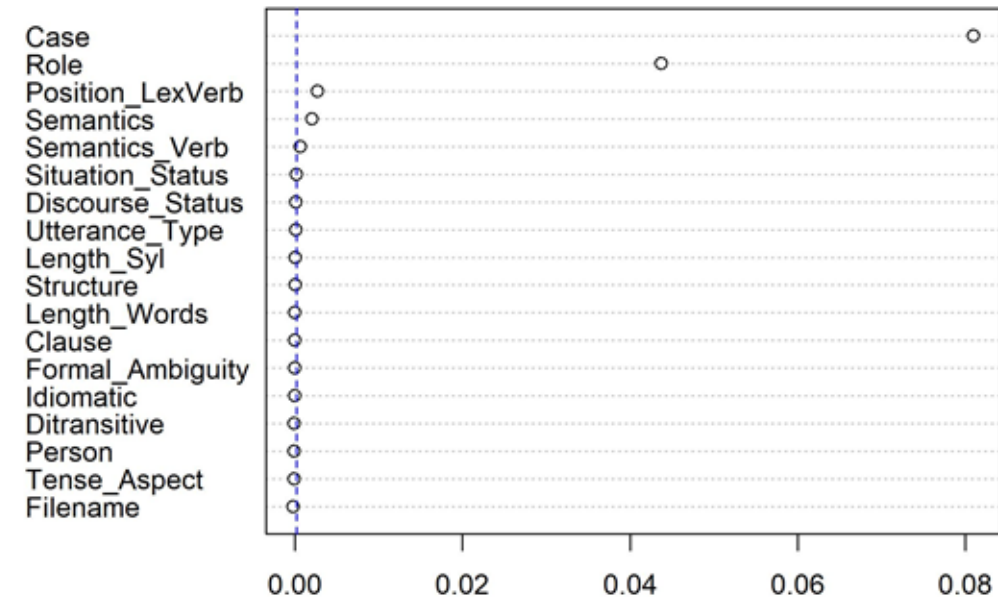
Conditional variable importance: Russian

Russian: cforest, Variable Importance, mtry = 10



Adult-directed speech

Russian CDS: cforest, Variable Importance, mtry = 10

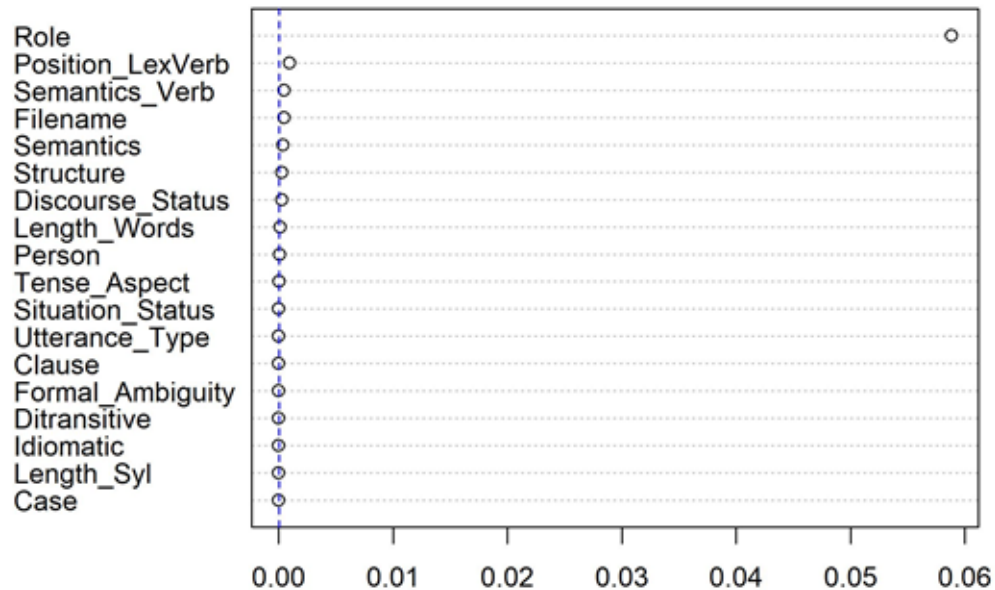


Child-directed speech

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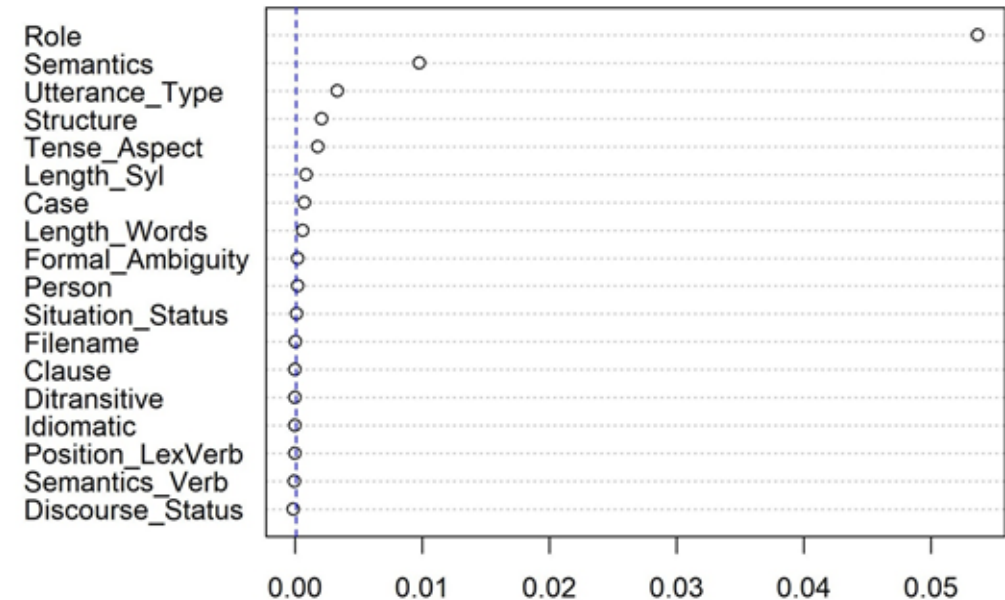
Conditional variable importance: Tagalog

Tagalog: cforest, Variable Importance, mtry = 10



Adult-directed speech

Tagalog CDS: cforest, Variable Importance, mtry = 10



Child-directed speech

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 - ~~Expectation: systematically.~~ **NO. There are too few examples where ambiguity is possible.**

Interpretation

- Discourse data provide no support for previous claims, at least, with regard to A and P arguments.
- Weak or no effect of givenness, in either form
 - Stronger effect of "Plan reuse" than previously assumed in flexible languages
- Very few cases of formal ambiguity:
 - High redundancy in the noisy communication channel (Gibson et al. 2013)

Outline

- Theoretical background: Communicative efficiency
- Conditional inference trees and random forests
- Illustration: Causative constructions of the world
- Hands-on case study with R: Order of A and P in three languages
- Discussion of your own research ideas, Q&A